

Direct Variation Word Problems

Algebra Worksheet · Grade 7–9

Name: _____

Date: _____

Learning Objectives

- Identify direct variation relationships from verbal descriptions and write the equation $y = kx$
- Find the constant of variation (k) given a pair of values
- Use the direct variation equation to solve for unknown quantities in real-world contexts

Problems

1. If y varies directly as x , and $y = 12$ when $x = 3$, find the constant of variation k .

$$y = kx$$

2. If y varies directly as x , and $y = 20$ when $x = 5$, find y when $x = 9$.

$$y = kx$$

3. The number of cupcakes you can bake varies directly with the amount of flour you have. You can bake 24 cupcakes with 3 cups of flour. How many cupcakes can you bake with 7 cups of flour?

$$c = kf$$

4. The amount you earn varies directly with the number of hours you work. You earn \$45 for working 5 hours. How much will you earn if you work 8 hours?

$$E = kh$$

5. If y varies directly as x , and $y = 48$ when $x = 6$, find y when $x = 15$.

$$y = kx$$

6. A car uses fuel that varies directly with the distance traveled. A car uses 4 liters of fuel to travel 60 km.



$$F = kd$$

7. The table below shows values of x and y . Determine whether the data represents a direct variation. If it does, state the constant of variation and write the equation.

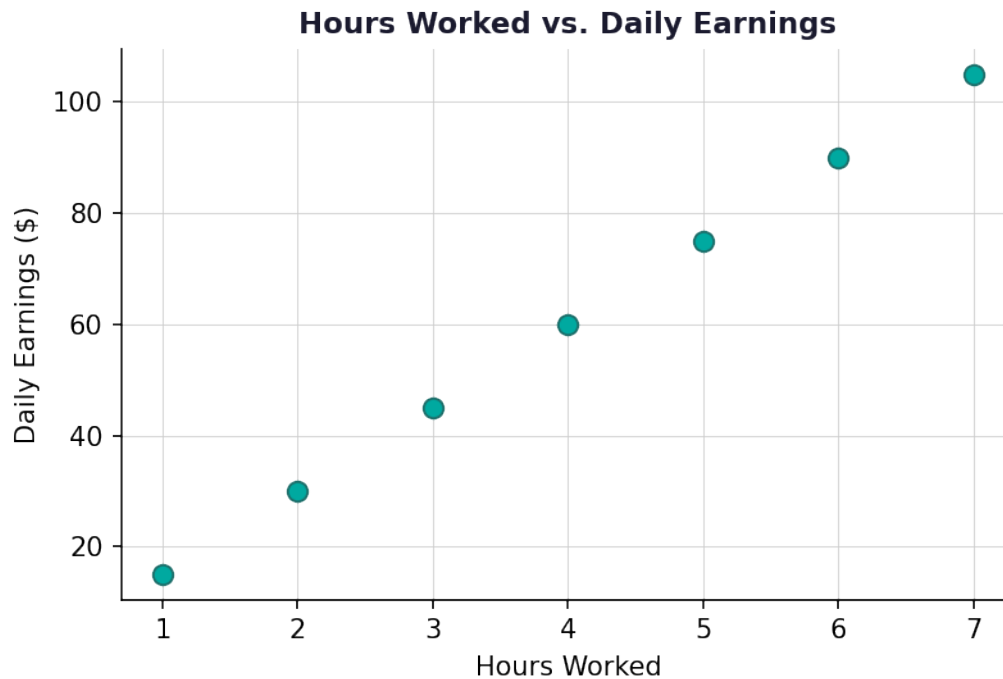
x	y	y/x
2	10	
4	20	
6	30	
8	40	

8. The distance a spring stretches varies directly with the force applied. A force of 12 Newtons stretches the spring 8 cm. What force is needed to stretch the spring 22.5 cm?

$$d = kF$$

9. The scatter plot below shows the relationship between hours worked per day and daily earnings for several employees. Based on the graph, does the data suggest a direct variation? Identify the approximate constant of variation from the data points.





10. At a factory, the number of units produced varies directly with the number of workers. With 8 workers, the factory produces 200 units per day. The factory wants to increase daily production to 325 units. However, each additional worker costs \$80 per day. If the current daily wage cost for 8 workers is \$480, what will be the new total daily wage cost after hiring the additional workers needed to meet the production goal?

$$u = kw$$



Direct Variation Word Problems — Answer Key

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Answer Key

1. Answer: $k = 4$

- Use the formula $y = kx$
- Substitute $y = 12$ and $x = 3$: $12 = k(3)$
- Divide both sides by 3: $k = 12 \div 3 = 4$

2. Answer: $y = 36$

- Find k : $k = y/x = 20/5 = 4$
- Write the equation: $y = 4x$
- Substitute $x = 9$: $y = 4(9) = 36$

3. Answer: 56 cupcakes

- Let c = number of cupcakes and f = cups of flour
- Find k : $k = 24/3 = 8$
- Write the equation: $c = 8f$
- Substitute $f = 7$: $c = 8(7) = 56$ cupcakes

4. Answer: \$72

- Let E = earnings and h = hours worked
- Find k : $k = 45/5 = 9$
- Write the equation: $E = 9h$
- Substitute $h = 8$: $E = 9(8) = \$72$

5. Answer: $y = 120$

- Find k : $k = 48/6 = 8$
- Write the equation: $y = 8x$
- Substitute $x = 15$: $y = 8(15) = 120$

6. Answer: 15 liters

- Let F = fuel (liters) and d = distance (km)
- Find k : $k = 4/60 = 1/15$
- Write the equation: $F = (1/15)d$
- Substitute $d = 225$: $F = 225/15 = 15$ liters

7. Answer: Yes, direct variation; $k = 5$; $y = 5x$

x	y	y/x
2	10	5
4	20	5



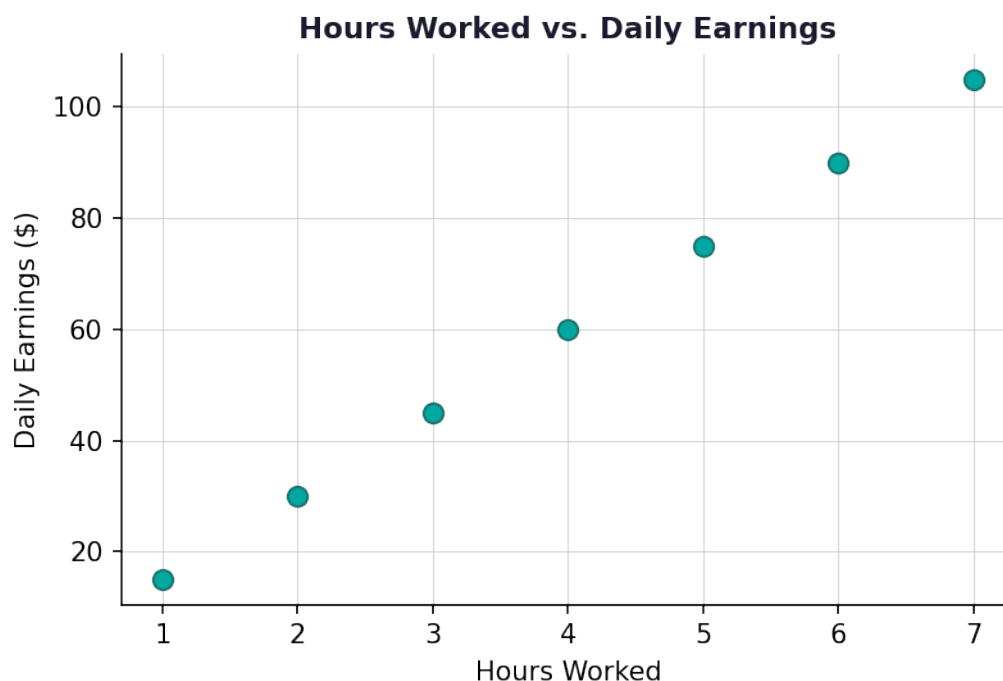
x	y	y/x
6	30	5
8	40	5

- Calculate y/x for each row: $10/2 = 5$, $20/4 = 5$, $30/6 = 5$, $40/8 = 5$
- Since y/x is constant, this is a direct variation
- The constant of variation is $k = 5$
- The equation is $y = 5x$

8. Answer: 33.75 Newtons

- Let d = distance stretched (cm) and F = force (Newtons)
- Find k : $k = d/F = 8/12 = 2/3$
- Write the equation: $d = (2/3)F$
- Substitute $d = 22.5$: $22.5 = (2/3)F$
- Solve for F : $F = 22.5 \times (3/2) = 33.75$ Newtons

9. Answer: Yes, direct variation; $k = 15$ (earning \$15 per hour)



- Check whether the points lie on a straight line through the origin
- Calculate y/x for each point: $15/1=15$, $30/2=15$, $45/3=15$, $60/4=15$, $75/5=15$, $90/6=15$, $105/7=15$
- Since $y/x = 15$ is constant and the line passes through $(0, 0)$, it is a direct variation
- The constant of variation $k = 15$, meaning each employee earns \$15 per hour
- The equation is $E = 15h$

10. Answer: New total daily wage cost = \$1,040

- Let u = units produced and w = number of workers





- Find k : $k = 200/8 = 25$ (units per worker)
 - Write the equation: $u = 25w$
 - Find the number of workers needed for 325 units: $325 = 25w \rightarrow w = 13$ workers
 - Additional workers needed: $13 - 8 = 5$ additional workers
 - Additional cost: $5 \times \$80 = \400 per day
 - New total daily wage cost: $\$480 + \$400 = \$1,040$
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